AI Image Generator

8. Performance

Version 1.0
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Platform: NVIDIA GeForce RTX 3090, Windows 11, Python 3.12

Stable Diffusion XL + Real-ESRGAN, served locally with Gradio, with per-image GPU telemetry recorded in SQLite.

8. Performance

All figures were measured on the development machine (RTX 3090, 24 GB, driver 610.88, torch 2.14.1 with CUDA 12.6) and taken from generations.db.

8.1 Generation (native size)

Size	Steps	Inference	Steps/s	Peak VRAM	Temp after	Power after
1024x1024	15	4.7 to 4.9 s	3.1 to 3.2	10.5 GB	55 to 56 C	377 to 379 W
1024x1024	30	9.7 s	3.1	10.5 GB	-	-
1344x768	40	12.8 to 13.5 s	3.0 to 3.1	10.4 GB	52 to 54 C	349 to 357 W
768x1344	40	11.2 to 11.4 s	3.5 to 3.6	10.5 GB	55 to 57 C	382 to 385 W

Throughput is close to 3.1 steps per second at 1 megapixel regardless of aspect ratio. Inference time is therefore almost exactly proportional to the step count: about 0.32 s per step.

8.2 Upscaling

From	To	Passes	Time	Output file
1344x768	3840x2194 (4K)	1	3.4 s	9.9 MB
768x1344	2194x3840 (4K)	1	2.9 s	7.9 MB
768x1344	4388x7680 (8K)	2	8.4 to 8.5 s	30.5 MB

A standalone first call measured 9.4 s for the 4K case because it included loading the weights; inside the app the weights are already resident.

8.3 End-to-end

Request	Total wall time (API call to response)
16:9, 40 steps, 4K	20.6 s, of which 13.5 s inference, 3.4 s upscale, remainder PNG encoding and transfer
1024x1024, 25 steps, no upscale	9.9 s

8.4 Memory

Stage	VRAM
Idle with SDXL loaded	about 7 GB allocated, 15.7 GB reserved by the allocator
Peak during 1 MP generation	10.4 to 10.5 GB
Upscaler weights	67 MB (half precision)
Peak during tiled upscale	no measurable increase over the generation peak

The 15.7 GB "used" figure that nvidia-smi reports while idle is allocator reservation, not live tensors; it is reused rather than growing.

8.5 Startup

Step	Time
First ever start (downloads 6.7 GB)	bandwidth dependent, several minutes
SDXL load from local cache	27 to 43 s (disk cache dependent)
Upscaler load	under 1 s

8.6 Aggregate over the session

Metric	Value
Images recorded	7
Total GPU inference time	71 s
Average inference	10.0 s
Upscaled images	4
Total upscale time	23.2 s
Hottest reading	57 C
Peak board power	385 W of a 390 W limit

8.7 Tuning levers

- Steps: the dominant cost. 15 steps gives usable drafts in under 5 s; 40 steps is the quality default for 4K.
- Count: images in a batch are generated sequentially, so time scales linearly.
- 8K: costs about 5 s more than 4K and produces 30 MB files; the browser preview is noticeably slower to load.
- Cards with less VRAM: generate.py enables CPU offload below 12 GB, which roughly doubles inference time but keeps the pipeline working.